

# JAZZSAMBA: A SYNCHRONOUS AND ASYNCHRONOUS MULTI-TAKE BAND AUDIO DATASET OF JAZZ STANDARDS FOR LIVE MUSIC MODELS

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## ABSTRACT

Machine learning has made strong progress on music tasks, both as assistive tools and as creative partners. However, most systems train on multitrack corpora that emphasize pop and rock. Jazz, with improvisation at the core of its practice, still lacks a well-annotated corpus of clean per-stem combo recordings on standards. We introduce **JazzSAMBA** (Jazz Synchronous and Asynchronous Multi-take Band Audio) to fill this gap: the first originally recorded jazz-combo multi-track dataset of standards with asynchronous (overdubbed) and synchronous (live ensemble) protocols, preferred and alternate takes chosen by the musicians, and timed annotations for bars, chords, sections, and soloists. JazzSAMBA covers 76 standards by eight musicians on drums, bass, piano, trumpet, and saxophone, with per-stem audio, mixtures, and MIDI. It can support chart-conditioned accompaniment, combo source separation, and form-aware music information retrieval. We demonstrate the dataset on two tasks: a jazz combo source-separation baseline and a chart-conditioned accompaniment ablation. The dataset, code, and samples are linked from the project demo page.<sup>1</sup>

**Index Terms**— jazz, multi-track audio, music information retrieval, dataset, music generation

## 1. INTRODUCTION

Machine learning has made strong progress on music understanding, live music generation, and other related music tasks. However, most systems train on multitrack corpora that emphasize pop and rock [1, 2, 3]. As a result, these datasets usually limit tasks to standard pop and rock instruments and styles, which usually involve compositions and arrangements created beforehand. Jazz, on the other hand, combines pre-composed charts with improvisation, so the improvisation space is bounded by harmony and structure rather than left unconstrained. Prior jazz resources cover audio-aligned harmony, formal structure, solo transcription, and solo-piano or stem-transcribed performances [4, 5, 6, 7], yet they rarely provide originally recorded multi-instrument combo stems of

standard repertoire [8]. However, interactive accompaniment, real-time generation, and source separation need corpora with clean ensemble stems. Chart-conditioned generation and form-aware analysis further benefit when those stems are paired with structural annotations that models can condition on or score against.

We propose JazzSAMBA, the first jazz-standard dataset with both clean per-stem combo recordings and performance-aligned chord and section labels. It provides asynchronous (overdubbed) and synchronous (live ensemble) protocols, preferred and alternate takes chosen by the musicians, and timed bar and soloist annotations. It comprises 76 standards by eight musicians on drums, bass, piano, and horns. These assets support chart-conditioned accompaniment, combo-standard instrument source separation, and form-aware MIR. We demonstrate two uses that depend on these assets: a jazz combo source-separation baseline, and a chart-conditioned causal accompaniment benchmark that ablates timed sections and chords against listen-only conditioning. The dataset, code, and samples are linked from the project demo page.<sup>1</sup>

## 2. RELATED WORK

**Table 1.** JazzSAMBA vs. selected related corpora.

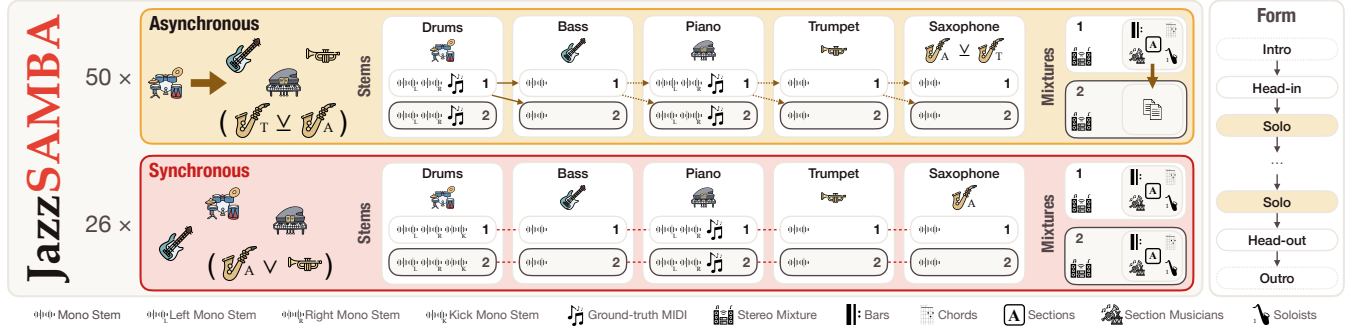
| Corpus                     | Stems          | Takes | Dual | Chords | Sections | Soloists | MIDI |
|----------------------------|----------------|-------|------|--------|----------|----------|------|
| MUSDB18 [1]                | ✓              |       |      |        |          |          |      |
| MoisesDB [2]               | ✓              |       |      |        |          |          |      |
| Slakh2100 [9]              | ✓              |       |      |        |          |          | ✓    |
| Jazz Trio Database [8]     | ✓ <sup>†</sup> |       |      |        |          |          | ✓*   |
| Jazz Harmony [4]           |                |       |      | ✓      | ✓        |          |      |
| Weimar Jazz Database [6]   |                |       |      | ✓      | ✓        | ✓        | ✓*   |
| Jazz Structure Dataset [5] |                |       |      |        | ✓        | ✓        |      |
| RWC Jazz [10]              |                |       |      |        | ✓        |          | ✓    |
| JAZZVAR [7]                |                |       |      | ✓      |          |          | ✓*   |
| H2H Music Improv [11]      | ✓              | ✓     |      |        |          |          |      |
| <b>JazzSAMBA</b>           | ✓              | ✓     | ✓    | ✓      | ✓        | ✓        | ✓    |

Dual: asynchronous and synchronous protocols. <sup>†</sup> Separation-derived stems; \* MIDI from automatic or monophonic transcription.

**Multitrack datasets.** Studio and synthetic multitracks (MUSDB18, MedleyDB, MoisesDB, Slakh2100) support separation and transcription outside jazz [1, 3, 2, 9]. JazzSAMBA targets that gap with originally recorded jazz-combo stems of standards rather than pop or rendered MIDI.

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<sup>1</sup><https://pnlong.github.io/jazzsamba-demo/>



**Fig. 1.** JazzSAMBA overview (50 asynchronous / 26 synchronous songs; about 8.1 h preferred-take, about 16.1 h both tiers). Labels 1 and 2 are take tiers. Asynchronous takes overdub in sequence (drums first; dotted arrows); synchronous takes record the whole ensemble together (dashed lines). Preferred mixtures are annotated for bars, chords, sections, and soloists; alternate async annotations are copied, while sync takes are labeled separately.

**Jazz corpora.** Table 1 compares related jazz and improvisation audio corpora along stems, takes, recording protocols, annotations, and MIDI. Prior jazz resources emphasize harmony and form labels, solo transcription, or solo piano without originally recorded multi-instrument combo stems [4, 5, 6, 7, 10]; where MIDI is present, it is often automatic or monophonic, and the jazz trio database supplies stems only via source separation [8]. H2H Music Improv provides multi-take free-improvisation per-stem audio with bidirectional intention annotations [11], but it is non-idiomatic and does not constrain the corpus to any specific style. JazzSAMBA unifies close-mic combo stems, dual protocols and takes, and timed chart annotations on standards.

**Interactive accompaniment.** Offline models include the Anticipatory Music Transformer [12]; online systems include RL-Duet, RealChords, StreamGen, RealJam, and Live-Band [13, 14, 15, 16, 17], mostly on symbolic pop/classical or non-jazz audio. Jazz suits such models: improvisation on shared charts couples freedom with form models can follow. JazzSAMBA could expand these models to real combo scenarios with its clean stems and chord and section annotations.

### 3. JAZZSAMBA DATASET

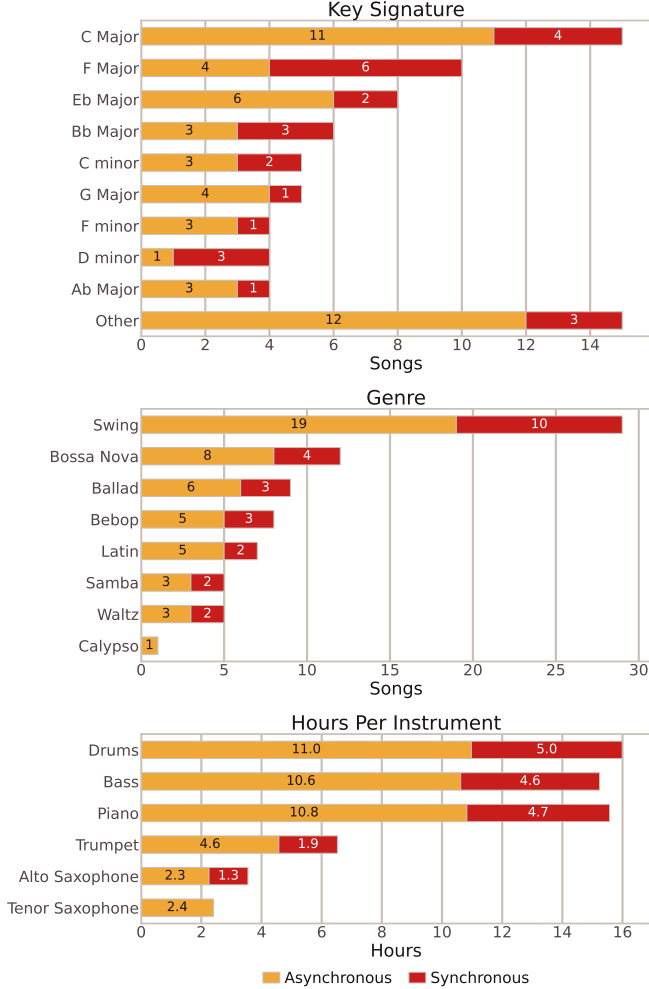
JazzSAMBA uses two recording protocols: instruments tracked one at a time, and the full band tracked together (Figure 1; Table 2). In the *asynchronous* protocol, musicians record separately: drums first establish a rhythmic foundation, then later instruments overdub on preferred earlier takes. Each instrument records two clean takes; preferred stems form the preferred mixture and the remaining takes form the alternate tier. Fixed-tempo overdubs let us annotate the preferred mixture once and copy those labels onto the alternate tier. In the *synchronous* protocol, the full band records two ensemble takes, musicians pick one preferred take, and close-mic bleed makes this the harder condition; takes are annotated independently.

**Table 2.** Asynchronous vs. synchronous protocol contents. Shared marks preferred labels copied to the alternate tier. GT denotes ground-truth; remaining MIDI is from MuScriptor AMT [18]. For stems with bleed, we also ship a Wiener-style de-bleeded version [19]. All audio is 48 kHz.

| Protocol     | Stems |       | Annotations |        | GT MIDI |       |
|--------------|-------|-------|-------------|--------|---------|-------|
|              | Clean | Bleed | Timed       | Shared | Drums   | Piano |
| Asynchronous | ✓     |       | ✓           | ✓      | ✓       | ✓     |
| Synchronous  |       | ✓     | ✓           |        |         | ✓     |

Both protocols store the same timed annotations on a shared bar grid. Bars give onset times and measure indices; chords, sections, and soloists use bar and fraction offsets so harmony and form stay aligned to the performance. Chord labels come from written lead sheets, not audio estimates: MusPyExpress [20] parses an internal chart collection, then musicians time-align them, keeping the written symbol and fields such as root, triad, seventh, extensions, alterations, and slash bass (e.g. *E-7b5*). Section labels use jazz form tags (intro, head-in, head-out, solos, outro), with nested letters when needed (e.g. *head-in/A*). A per-section musician table records who plays; soloist rows give an ordered solo sequence.

The corpus covers 76 standards by eight musicians on drums, bass, piano, trumpet, and saxophone (alto or tenor, exclusive per song): 50 asynchronous and 26 synchronous. Repertoire spans swing, bossa nova, ballad, bebop, and latin feels, mostly in 4/4 with both swung and straight grooves. Preferred mixtures total about 8.1 h (5.5 h asynchronous, 2.5 h synchronous); both tiers reach about 16.1 h at 75–270 BPM. Figure 2 summarizes keys, genres, and hours per instrument; listening examples are on the demo page.



**Fig. 2.** JazzSAMBA corpus statistics by protocol (asynchronous or synchronous): key signatures, genres, and active hours per instrument from section musician annotations over both take tiers, excluding silence while sitting out.

## 4. EXPERIMENTS

We demonstrate JazzSAMBA in two settings: source separation and chart-conditioned accompaniment. Source separation fine-tunes and evaluates on asynchronous stem-sum mixtures for drums, bass, piano, and horns. Chart-conditioned accompaniment trains on both protocols and ablates oracle-timed chord and section masks against listen-only conditioning on asynchronous and synchronous test sets.

### 4.1. Source separation

Standard music source-separation models typically target vocals, drums, bass, and other, collapsing instruments outside that taxonomy into a catch-all stem. JazzSAMBA enables fine-tuning separators for jazz-combo stems like piano and

**Table 3.** Asynchronous-test SI-SDR  $\uparrow$  (dB), full-track ( $n=5$ ). Dashes mark unfair zero-shot or ChoraleBricks cells (no matching stem supervision). We bold the best reportable score within each model.

| Model          | FT            | Drums        | Bass         | Piano        | Horns        |
|----------------|---------------|--------------|--------------|--------------|--------------|
| HT-Demucs 6s   | —             | 13.01        | 15.07        | 4.92         | —            |
|                | JazzSAMBA     | <b>15.09</b> | <b>17.03</b> | <b>9.42</b>  | <b>14.62</b> |
|                | ChoraleBricks | —            | 15.02        | —            | 8.98         |
| Open-Unmix     | —             | 5.24         | 5.63         | —            | —            |
|                | JazzSAMBA     | <b>8.37</b>  | <b>9.74</b>  | <b>6.07</b>  | <b>9.38</b>  |
|                | ChoraleBricks | —            | 5.63         | —            | −0.66        |
| BS-RoFormer SW | —             | <b>19.12</b> | <b>20.83</b> | <b>18.31</b> | <b>21.57</b> |
|                | JazzSAMBA     | 18.97        | 20.62        | 17.54        | 21.15        |
|                | ChoraleBricks | —            | 17.59        | —            | 2.00         |

horns. We fine-tune HT-Demucs 6s [21], a hybrid waveform-spectrogram Transformer with a dedicated piano stem; Open-Unmix [22], a spectrogram-masking LSTM baseline; and BS-RoFormer SW [23], a band-split RoPE Transformer, on asynchronous JazzSAMBA takes to separate stem-sum mixtures of drums, bass, piano, and horns. Horns are initialized from the pretrained “other” stem; Open-Unmix has no piano stem, so we initialize its piano head from a copy of pretrained “other”. We compare zero-shot against fine-tuning on JazzSAMBA or ChoraleBricks [24], which provides isolated wind parts including flute, oboe, clarinet, trumpet, saxophone, baritone, trombone, and tuba, to test whether wind stems transfer. Training uses shared mix-and-separate 4s crops: we RMS-normalize each stem, apply a random per-stem gain, sum to a mixture, then peak-normalize the mixture and matched targets together. We report full-track SI-SDR [25], which measures how closely an estimate matches a reference while ignoring overall loudness, on the asynchronous test set ( $n=5$ ) with early stopping on validation loss (Table 3).

Zero-shot BS-RoFormer is strongest on reportable stems; JazzSAMBA fine-tuning keeps it within about 0.8 dB, so zero-shot remains the headline BS result. JazzSAMBA fine-tuning makes horns reportable for HT-Demucs and piano and horns reportable for Open-Unmix, and it also improves drums and bass on those models. ChoraleBricks fine-tuning does not improve JazzSAMBA test scores (e.g., BS horns 21.57→2.00 dB), suggesting limited transfer from wind chorales to this jazz-combo setting.

### 4.2. Chart-conditioned accompaniment

We demonstrate how researchers may benchmark live music accompaniment using JazzSAMBA by comparing chart conditioning against listen-only baselines. We fine-tune Stream-Gen [15], a streaming latent model for stem-held-out accompaniment, with a causal decoder ( $t_f=0$ ) so each frame depends only on past mix audio and optional chart features, suited to live accompaniment. A four-way ablation over

**Table 4.** Accompaniment scores on 60 s free-run from form landmarks (asynchronous test  $n=5$ , synchronous test  $n=3$ ; mean over landmarks). D, B, P, and H denote drums, bass, piano, and horns. Ground Truth scores the annotated target stem against the condition mix. We bold the best score per column among chart conditions (excluding Ground Truth).

| Protocol     | Condition    | COCOLA $\uparrow$ |             |             |             | Beat F1 $\uparrow$ |             |             |             |
|--------------|--------------|-------------------|-------------|-------------|-------------|--------------------|-------------|-------------|-------------|
|              |              | D                 | B           | P           | H           | D                  | B           | P           | H           |
| Asynchronous | Ground Truth | 55.6              | 62.1        | 63.4        | 64.5        | 1.00               | 0.80        | 0.48        | 0.29        |
|              | Listen-Only  | 52.2              | 60.4        | 59.8        | 61.4        | 0.23               | <b>0.16</b> | 0.13        | 0.02        |
|              | +Sections    | 50.7              | 59.8        | 59.7        | 61.9        | 0.29               | 0.09        | 0.13        | 0.05        |
|              | +Chords      | <b>52.5</b>       | <b>60.6</b> | <b>60.0</b> | <b>62.3</b> | 0.30               | 0.09        | <b>0.15</b> | <b>0.06</b> |
|              | +Both        | 51.4              | 60.4        | 59.9        | 61.7        | <b>0.34</b>        | 0.10        | 0.12        | 0.04        |
| Synchronous  | Ground Truth | 58.4              | 61.2        | 62.9        | 65.9        | 0.97               | 0.74        | 0.67        | 0.78        |
|              | Listen-Only  | 53.2              | <b>60.3</b> | 58.6        | 60.8        | 0.26               | <b>0.12</b> | 0.09        | 0.05        |
|              | +Sections    | 52.8              | 58.9        | 58.3        | <b>61.4</b> | 0.27               | 0.06        | 0.14        | 0.01        |
|              | +Chords      | <b>53.5</b>       | 59.1        | <b>58.9</b> | 60.9        | 0.27               | 0.09        | 0.09        | <b>0.05</b> |
|              | +Both        | 51.3              | 59.9        | 58.7        | 60.9        | <b>0.31</b>        | 0.07        | <b>0.15</b> | 0.00        |

listen-only, +sections, +chords, and +both trains specialists on both protocols with 10 s crops and optional oracle-timed JazzSAMBA charts. At inference we free-run 60 s from up to three form landmarks per take (head-in and the first two soloist entries), conditioning only on the non-target mix and chart. Table 4 reports COCOLA [26] and Beat Alignment F1 [17, 27] as mean over landmarks, for asynchronous ( $n=5$ ) and synchronous ( $n=3$ ) test songs. COCOLA scores contrastive coherence between the non-target condition mix and the generated stem; Beat Alignment F1 compares an oracle bar-grid metronome from the take annotations to beats detected on the generated stem.

Across both protocols, chart masks change Beat F1 more than COCOLA: Beat F1 swings farther across conditions and the best mask often changes by stem, while +chords often edges StreamGen’s COCOLA over listen-only, showing that JazzSAMBA’s annotations can discriminate conditioning strategies. Synchronous Beat F1 is somewhat lower under live bleed, while COCOLA remains comparable to the asynchronous setting. Live audio accompaniment remains a difficult task; these results are one baseline where JazzSAMBA’s stems and annotations help.

## 5. CONCLUSION

JazzSAMBA is the first jazz-standard dataset with per-player stems and annotations for chords, sections, and soloists. It includes 76 standards with multi-take audio and MIDI in asynchronous and synchronous settings, performed by eight musicians. Paired with performance-aligned charts, it supports music information retrieval and generation. We show that it enables jazz-combo source separation for piano and horns, and benchmarks live chart-conditioned accompaniment. We release JazzSAMBA hoping others will push beyond these

baselines toward interactive chart-following accompaniment, form-aware combo analysis, and models that handle take-to-take and live-ensemble variation.

## 6. COMPLIANCE WITH ETHICAL STANDARDS

Performing musicians are credited as coauthors and consented to the recording and public release of the dataset. Recordings are original studio performances. The dataset contains those recordings and our annotations only.

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